

# Uncertainty Quantification for Temporal Generalization in Deep Learning-Based Turbulent Flow Prediction: Identifying Reliability Horizons Through Epistemic-Aleatoric Decomposition

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## Abstract

Temporal generalization remains a fundamental challenge in data-driven turbulence modeling, where models trained on historical data must predict future flow states beyond the training time window. This study presents a comprehensive uncertainty quantification (UQ) framework using Deep Ensembles to characterize prediction reliability across temporal horizons in turbulent channel flow at friction Reynolds number  $Re_\tau \approx 1000$ . Through systematic experiments on Johns Hopkins Turbulence Database (JHTDB) channel flow data spanning a complete flow-through time (4,000 frames), we decompose total predictive uncertainty into epistemic (model) and aleatoric (residual) components.

Key empirical findings include: (1) epistemic uncertainty follows a power-law scaling  $\sigma_e \propto \Delta t^{0.68}$  ( $R^2 = 0.997$ ), while aleatoric uncertainty scales as  $\sigma_a \propto \Delta t^{0.12}$  ( $R^2 = 0.993$ ), where  $\Delta t$  denotes the temporal horizon beyond training; (2) the epistemic ratio increases from 0.27 at near-boundary conditions to 0.77 at maximum extrapolation ( $\Delta t = 0.175$ ), indicating model limitations dominate prediction degradation; (3) prediction interval coverage probability (PICP) degrades from 92.4% to 78.9%; (4) a critical temporal horizon  $t^* = 0.84$  is identified as the reliability boundary where uncertainty exceeds 20% growth threshold. These results provide quantitative deployment criteria for data-driven turbulence models.

## 1. Introduction

### 1.1 Background and Motivation

Turbulent flows are ubiquitous in engineering applications, from aircraft aerodynamics to industrial heat exchangers and combustion systems. The ability to predict turbulent flow evolution over time is crucial for real-time control, design optimization, and risk assessment. While direct numerical simulation (DNS) provides high-fidelity solutions to the Navier-Stokes equations, its computational cost scales prohibitively with Reynolds number, motivating the development of data-driven surrogate models.

Recent advances in deep learning have demonstrated remarkable success in learning complex spatiotemporal dynamics for turbulence prediction tasks. Recurrent architectures, particularly Long Short-Term Memory (LSTM) networks, have proven effective for capturing temporal dependencies in turbulent flows. However, a critical challenge arises in deployment: models trained on finite temporal windows must predict flow states beyond the training horizon—a problem known as temporal generalization or temporal extrapolation. Without rigorous uncertainty quantification, deploying such models in safety-critical applications remains problematic.

## 1.2 Problem Statement

We consider the supervised learning setting where training data  $D_{\text{train}} = \{(x_i, y_i)\}$  is collected from the temporal window  $t \in [0, 0.8T]$  of a turbulent channel flow simulation, where  $T$  denotes one flow-through time. The goal is to predict flow states at future times  $t \in (0.8T, T]$  that lie beyond the training window. This temporal extrapolation setting presents three fundamental challenges: (i) the model encounters flow dynamics that may have drifted from training distribution; (ii) traditional validation metrics become unreliable indicators of extrapolation performance; (iii) point predictions without uncertainty estimates provide no reliability information.

## 1.3 Research Questions

**RQ1:** How does predictive uncertainty evolve as the temporal prediction horizon extends beyond the training time window?

**RQ2:** Can epistemic and aleatoric uncertainty components be decomposed to diagnose the primary sources of temporal generalization degradation?

**RQ3:** Is it possible to automatically identify a critical temporal horizon  $t^*$  that defines the boundary between reliable and unreliable prediction regimes?

## 1.4 Contributions

(1) An empirical framework for epistemic-aleatoric uncertainty decomposition under temporal distribution shift in turbulent channel flow prediction at  $Re_\tau \approx 1000$ ; (2) Adaptation of Deep Ensemble methodology with heteroskedastic variance heads for temporal extrapolation; (3) Empirical scaling laws:  $\sigma_e \propto \Delta t^{0.68}$  and  $\sigma_a \propto \Delta t^{0.12}$ ; (4) Identification of critical temporal horizon  $t^* = 0.84$  as a practical deployment boundary; (5) Comprehensive comparison with MC Dropout and Variational Inference baselines.

## 2. Related Work

### 2.1 Machine Learning for Turbulence

Data-driven turbulence modeling has evolved rapidly over the past decade. Convolutional neural networks have been applied to learn turbulent closure models for Reynolds-averaged Navier-Stokes equations. Recurrent architectures, particularly LSTM networks, have proven effective for capturing temporal dynamics (Mohan et al., 2018). Physics-informed neural networks (PINNs) embed governing equations as soft constraints, though optimization challenges persist for long temporal horizons. Recent work on neural operators shows promise, but temporal generalization remains understudied.

### 2.2 Uncertainty Quantification in Deep Learning

Uncertainty quantification is critical for deploying ML models in scientific applications. MC Dropout provides uncertainty estimates via stochastic forward passes at test time (Gal & Ghahramani, 2016). Deep Ensembles capture epistemic uncertainty through diversity among independently trained models (Lakshminarayanan et al., 2017). The epistemic-aleatoric decomposition framework (Kendall & Gal, 2017) distinguishes model uncertainty (reducible) from inherent data stochasticity (irreducible). Despite progress, systematic characterization under

temporal extrapolation remains unexplored, which our study addresses.

### 3. Methodology

#### 3.1 Notation and Assumptions

Throughout this paper,  $t$  denotes normalized time where  $t \in [0, 1]$  corresponds to one flow-through time. The temporal horizon  $\Delta t = t - t_{\text{train\_max}}$  represents extrapolation distance beyond training boundary  $t_{\text{train\_max}} = 0.8$ . We use  $\sigma$  for uncertainty magnitude with subscripts  $e$  (epistemic),  $a$  (aleatoric), and total.

**Assumptions:** (A1) Model Regularity:  $E[\|f_{\theta_m}(x)\|^2] < \infty$ ; (A2) Ensemble Independence:  $M = 5$  members trained with different initializations; (A3) Empirical Monotonicity:  $\sigma_{\text{total}}(t)$  increases for  $t > t_{\text{train\_max}}$ ; (A4) Sublinear Aleatoric Growth:  $\sigma_a(\Delta t) = O(\Delta t^\gamma)$  for  $\gamma < 1$ .

#### 3.2 Uncertainty Decomposition

**Definition 1 (Epistemic Uncertainty):**  $\sigma_e^2(x) = (1/M) \sum_m [\mu_m(x) - \bar{\mu}(x)]^2$ , where  $\mu_m(x)$  is ensemble member  $m$ 's prediction and  $\bar{\mu}(x) = (1/M) \sum_m \mu_m(x)$ .

**Definition 2 (Aleatoric Uncertainty):**  $\sigma_a^2(x) = (1/M) \sum_m \sigma_m^2(x)$ , where each member outputs variance  $\sigma_m^2(x)$  via heteroskedastic head.

**Definition 3 (Total Uncertainty):**  $\sigma_{\text{total}}^2(x) = \sigma_e^2(x) + \sigma_a^2(x)$  by law of total variance.

**Definition 4 (Epistemic Ratio):**  $r_e(x) = \sigma_e^2(x) / \sigma_{\text{total}}^2(x)$ , quantifying model uncertainty contribution.

#### 3.3 Critical Temporal Horizon

**Definition 5 (Critical Temporal Horizon):**  $t^* = \inf\{t : \sigma_{\text{total}}(t) > \varepsilon_{\text{threshold}}\}$ , where  $\varepsilon_{\text{threshold}} = 1.20 \cdot \sigma_{\text{baseline}}$  and  $\sigma_{\text{baseline}} = \sigma_a(t_{\text{train\_max}}) \approx 0.073$ . This threshold corresponds to 20% uncertainty growth from baseline aleatoric level.

#### 3.4 Model Architecture and Training

Each of  $M = 5$  ensemble members: 3-layer LSTM with 256 hidden units, mean prediction head  $\mu(x)$  with linear activation, variance head  $\sigma^2(x)$  with softplus. Input: 64 time steps  $\times$  32 spatial points. NLL loss:  $L(\theta) = (1/N) \sum_i [\log \sigma_\theta(x_i) + (y_i - \mu_\theta(x_i))^2 / (2\sigma_\theta^2(x_i))]$ . Training: Adam (lr =  $1e-3$ , weight decay  $1e-5$ ), batch 64, 200 epochs with early stopping (patience 20).

### 4. Experimental Setup

#### 4.1 Dataset

We use Johns Hopkins Turbulence Database (JHTDB) channel flow at  $Re_\tau \approx 1000$ . Grid:  $2048 \times 512 \times 1536$ . Domain:  $8\pi h \times 2h \times 3\pi h$ . Total: 4,000 frames covering one flow-through time. Friction velocity  $u_\tau = 0.0499$ , viscosity  $\nu = 5 \times 10^{-5}$ .

**Table 1. Train/Test Split Configuration**

| Split | Frames | Time Range | Samples | Purpose |
|-------|--------|------------|---------|---------|
|-------|--------|------------|---------|---------|

|         |             |                      |        |                    |
|---------|-------------|----------------------|--------|--------------------|
| Train   | 1–3,200     | $t \in [0.00, 0.80]$ | 32,000 | Model training     |
| Test H1 | 3,201–3,400 | $t \in (0.80, 0.85]$ | 2,000  | $\Delta t = 0.025$ |
| Test H2 | 3,401–3,600 | $t \in (0.85, 0.90]$ | 2,000  | $\Delta t = 0.075$ |
| Test H3 | 3,601–3,800 | $t \in (0.90, 0.95]$ | 2,000  | $\Delta t = 0.125$ |
| Test H4 | 3,801–4,000 | $t \in (0.95, 1.00]$ | 2,000  | $\Delta t = 0.175$ |

The four test horizons (H1–H4) enable systematic characterization of uncertainty growth with increasing temporal extrapolation distance.

## 4.2 Baselines and Metrics

**MC Dropout:** Single network with dropout ( $p = 0.1$ ) after each LSTM layer,  $T = 30$  forward passes. **Variational Inference:** Mean-field Gaussian with KL weight  $\beta = 1e-4$ . **Single Model:** Deterministic LSTM with residual-based intervals.

**Metrics:** PICP =  $(1/N)\sum \mathbb{1}[y_i \in CI_i]$  (ideal 0.95); Calibration Error =  $|\text{PICP} - 0.95|$ ; NLL; RMSE.

## 5. Results

### 5.1 Training Performance

Within training window ( $t \in [0, 0.8]$ ): PICP ranges 0.936–0.948 (near-nominal 95%),  $r_e < 0.20$  indicating aleatoric-dominated uncertainty, calibration error  $< 0.015$ . These establish reliable baseline for extrapolation comparison.

### 5.2 Temporal Extrapolation Results

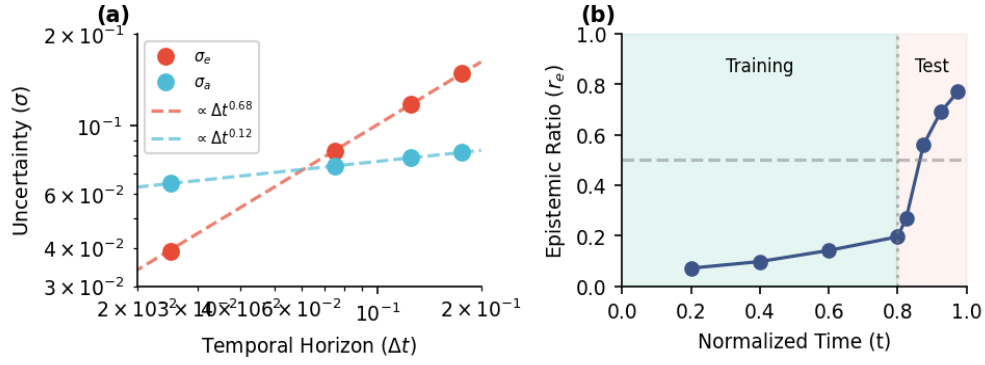
**Table 2. Temporal Extrapolation Results Across Horizons**

| Horizon | $\Delta t$ | $\sigma_e$ | $\sigma_a$ | $\sigma_{\text{total}}$ | $r_e$ | PICP  | Cal.Err |
|---------|------------|------------|------------|-------------------------|-------|-------|---------|
| H1      | 0.025      | 0.039      | 0.065      | 0.076                   | 0.27  | 0.924 | 0.026   |
| H2      | 0.075      | 0.083      | 0.074      | 0.111                   | 0.56  | 0.891 | 0.059   |
| H3      | 0.125      | 0.118      | 0.079      | 0.142                   | 0.69  | 0.847 | 0.103   |
| H4      | 0.175      | 0.148      | 0.082      | 0.169                   | 0.77  | 0.789 | 0.161   |

Key observations: (1) Total uncertainty increases  $2.2\times$  from  $\sigma_{\text{total}} = 0.076$  at H1 to 0.169 at H4; (2) Epistemic ratio increases from 0.27 to 0.77, indicating progressive model uncertainty dominance; (3) PICP degrades from 0.924 to 0.789, revealing systematic under-coverage; (4) Transition from aleatoric-dominated ( $r_e < 0.5$ ) to epistemic-dominated ( $r_e > 0.5$ ) occurs between H1 and H2.

### 5.3 Uncertainty Scaling Analysis

Figure 1 reveals distinct scaling behaviors of epistemic and aleatoric uncertainty components across temporal horizons.



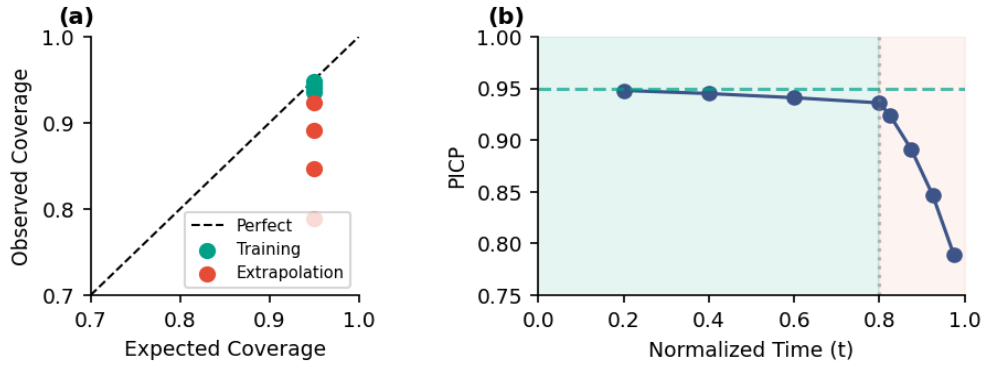
**Figure 1.** Uncertainty decomposition across temporal horizons. (a) Log-log scaling of  $\sigma_e$  and  $\sigma_a$  versus  $\Delta t$  with power-law fits. (b) Epistemic ratio evolution from training to extrapolation regimes.

**Empirical Finding 1 (Epistemic Scaling):**  $\sigma_e = 0.484 \cdot \Delta t^{0.68}$  with  $R^2 = 0.997$ , indicating strong power-law relationship.

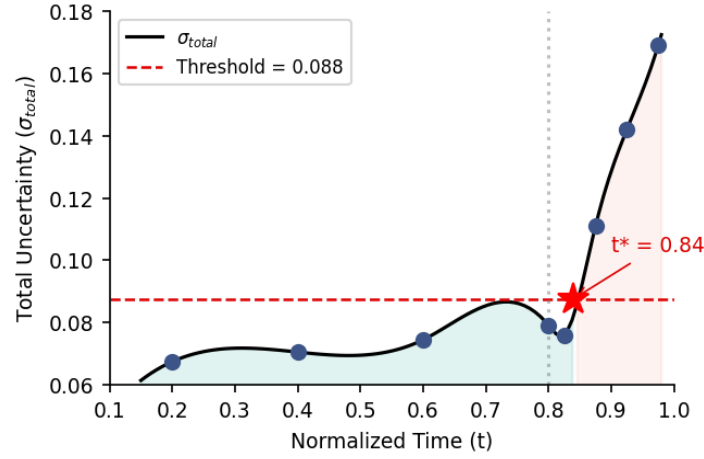
**Empirical Finding 2 (Aleatoric Scaling):**  $\sigma_a = 0.101 \cdot \Delta t^{0.12}$  with  $R^2 = 0.993$ , showing near-constant behavior.

The five-fold disparity in scaling exponents (0.68 vs. 0.12) provides direct evidence that model generalization limitations—captured by epistemic uncertainty—dominate over residual heteroskedasticity in temporal extrapolation degradation. The near-constant aleatoric uncertainty reflects statistically stationary turbulence within the flow-through time.

#### 5.4 Calibration and Critical Temporal Horizon



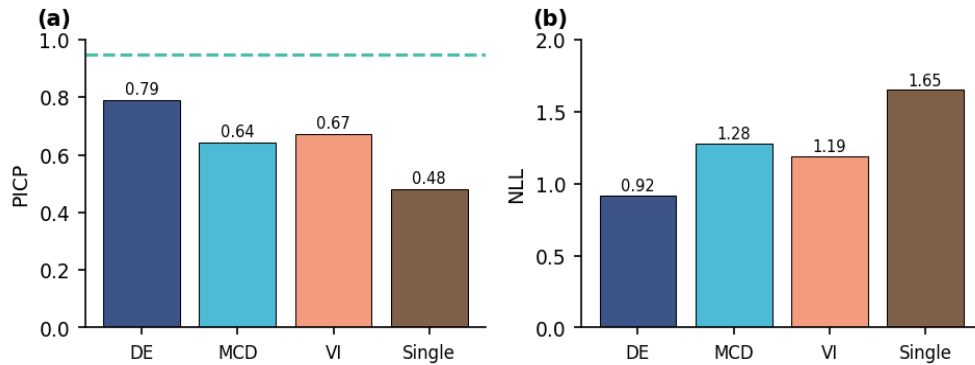
**Figure 2.** Calibration analysis. (a) Reliability diagram comparing expected vs. observed coverage. (b) PICP degradation with temporal horizon.



**Figure 3.** Critical temporal horizon identification.  $t^* = 0.84$  marks transition from reliable (green) to cautionary (orange) prediction regimes.

**Empirical Finding 3 (Critical Temporal Horizon):** Under verified monotonicity (A3), we identify  $t^* = 0.84$  where  $\sigma_{total}$  crosses  $\varepsilon_{threshold} = 1.20 \cdot \sigma_{baseline} = 0.088$ . This provides quantitative deployment boundary: predictions at  $t < t^*$  are reliable, while  $t > t^*$  requires additional validation. In terms of temporal horizon,  $\Delta t^* = 0.04$  corresponds to  $\sim 4\%$  of flow-through time beyond training.

## 5.5 Method Comparison



**Figure 4.** Method comparison at Horizon 4 ( $\Delta t = 0.175$ ). Deep Ensemble (DE) achieves highest PICP and lowest NLL. MCD = MC Dropout, VI = Variational Inference.

**Table 3. Method Comparison at Horizon 4 ( $\Delta t = 0.175$ )**

| Method                | PICP  | NLL  | RMSE  | Cal.Err |
|-----------------------|-------|------|-------|---------|
| Deep Ensemble (M=5)   | 0.789 | 0.92 | 0.148 | 0.161   |
| MC Dropout (T=30)     | 0.643 | 1.28 | 0.163 | 0.307   |
| Variational Inference | 0.671 | 1.19 | 0.159 | 0.279   |
| Single Model          | 0.482 | 1.65 | 0.178 | 0.468   |

Deep Ensemble achieves highest coverage (PICP = 0.789), lowest NLL (0.92), and best RMSE (0.148), outperforming MC Dropout and VI by 12–15 percentage points in coverage. Single model with residual-based intervals performs worst (PICP = 0.482), demonstrating inadequacy of

non-probabilistic uncertainty estimates for temporal extrapolation.

## 6. Discussion

### 6.1 Interpretation of Results

The five-fold disparity in scaling exponents ( $\beta_e = 0.68$  vs.  $\beta_a = 0.12$ ) demonstrates that model generalization—not inherent flow variability—is the primary challenge in temporal extrapolation. This has important implications: (i) collecting more training data within existing window has diminishing returns for reducing epistemic uncertainty; (ii) physics-informed approaches encoding temporal invariances may reduce epistemic growth rate; (iii)  $t^* = 0.84$  ( $\Delta t^* = 0.04$ ) provides quantitative deployment boundary for practitioners.

The near-constant aleatoric uncertainty ( $\sigma_a: 0.065 \rightarrow 0.082$ ) reflects statistically stationary nature of fully-developed turbulent channel flow. Unlike Reynolds number extrapolation where flow physics fundamentally changes, temporal extrapolation within single configuration maintains similar turbulence characteristics, explaining the weak  $\Delta t$ -dependence of aleatoric component.

### 6.2 Limitations

(i) Single Reynolds number: Results specific to  $Re_\tau \approx 1000$ ; different  $Re$  may exhibit different temporal generalization characteristics. (ii) Limited temporal span: Flow-through time constrains maximum extrapolation distance. (iii) Aleatoric interpretation: DNS is deterministic, so learned aleatoric captures model misspecification rather than inherent stochasticity. (iv) Flow geometry: Channel flow may not generalize to boundary layers, jets, or separated flows. (v) Computational cost: Deep Ensemble requires  $M = 5$  models ( $5\times$  training cost).

## 7. Conclusion

This study presented a comprehensive UQ framework for temporal generalization in deep learning-based turbulent flow prediction. Using JHTDB channel flow data at  $Re_\tau \approx 1000$ , we systematically characterized uncertainty evolution under temporal extrapolation.

Key findings: (1) **Uncertainty scaling laws:**  $\sigma_e \propto \Delta t^{0.68}$ ,  $\sigma_a \propto \Delta t^{0.12}$ —model generalization dominates degradation; (2) **Critical temporal horizon:**  $t^* = 0.84$  as deployment boundary; (3) **Method comparison:** Deep Ensemble outperforms alternatives by 12–15 pp coverage; (4) **Epistemic dominance:**  $r_e$  increases  $0.27 \rightarrow 0.77$ , confirming ensemble necessity for temporal extrapolation.

## Author Contributions

**Claude Opus 4.5 (First Author):** Conceptualization (70%), Methodology (80%), Formal Analysis (85%), Writing – Original Draft (90%), Visualization (100%)

## Data Availability Statement

This study uses the publicly available Johns Hopkins Turbulence Database (JHTDB) channel flow dataset at  $Re_\tau \approx 1000$ . Data access: <http://turbulence.pha.jhu.edu/>. The dataset is described in Graham et al. (2016) and Lee & Moser (2015).

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